

Stress Lab — Economic Model

How the layered shields keep a RoundFi pool solvent under default

RoundFi

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Derived document. This is a derivation of [docs/spec/MASTER-SPEC.md](#) (§4.3 Contribution split, §4.4 Stake & the Triple Shield, §4.5 Default settlement, §8 Economics). The Master Spec is the single source of truth; if a number here disagrees with it, the Master Spec wins. Every constant is pinned to the deployed Jun 2026 source (`programs/roundfi-core/` , `crates/math/`).

1. The thesis

A rotating savings pool (ROSCA) has one structural hazard: **someone draws the credit early and then stops paying**. They received the carta in cycle 2; they owe installments through cycle 8; they walk. In an informal *consórcio* that loss lands on the remaining members and the group dissolves. RoundFi's job is to make that default **recoverable** — the pool keeps paying its drawn members, the loss is absorbed by capital the defaulter themselves posted, and the event becomes a durable reputation signal rather than a dissolution.

This document is the **Stress Lab**: the solvency model and the invariants that make a default recoverable. It frames three things. First, that the pool is **self-funding** — no external subsidy is required for the cycle-0 draw to clear. Second, that a **Triple Shield** of layered capital, seized in a fixed order, covers a missed installment without touching honest members' float. Third, that the **stake discount** offered to proven members (50% → 3% of the credit) does **not** raise protocol risk, because the upper tiers are gated on a long completed -pool history that is itself the risk signal.

The whole model rests on one separation carried over from the reputation design: **receiving capital is not merit; keeping your obligations after receiving it is**. The shields collateralize that distinction; the self-funding guard makes it affordable.

2. The contribution split — three streams from one installment

Every `contribute` call splits the member's `installment` into three streams before any of it reaches the pool float. The split is enforced on-chain, not trusted off-chain.

Slice	bps	Destination	Role
Solidarity	100 (1%)	<code>solidarity_vault</code>	First tranche of any default cover
Escrow	<code>escrow_release_bps</code> (default 2500 = 25%)	<code>escrow_vault</code>	Vesting collateral / second tranche
Pool float	remainder (default 74%)	<code>pool_usdc_vault</code>	What pays the drawn member

Source: MASTER-SPEC §4.3; `crates/math/src/seed_draw.rs`.

The three percentages are not arbitrary — each maps to a defensive role. The **1% solidarity** slice is a small, communal, always-first buffer (the "Cofre Solidário"). The **25% escrow** slice is the member's own money, held back and released as they prove discipline (§4). The **74% float** is the working capital that actually funds the rotating draw. Note that the solidarity and escrow slices are taken **off the top**: only 74% of each installment is available to pay the person drawn this cycle. That retention is what the next section's guard is built around.

3. The Seed-Draw self-funding guard

The single most important solvency invariant is that **the very first draw can be paid out of contributions alone**. In cycle 0, members have each paid one installment; one member draws the full credit (`carta`). If the float retained from those installments is smaller than the credit, the pool is insolvent on day one — the cycle-0 claim under-runs the vault.

RoundFi refuses to create such a pool. `create_pool` evaluates the **Seed-Draw viability guard** before allocating, and rejects any geometry whose math doesn't close:

```
members × installment × (MAX_BPS - solidarity_bps - escrow_release_bps) /
MAX_BPS ≥ credit
```

With the default split this is the headline form from §8.1:

```
members × installment × 0.74 ≥ credit
```

Source: MASTER-SPEC §4.3 / §8.1; `crates/math/src/seed_draw.rs`. A pool that fails the guard reverts with `PoolNotViable` at creation time, so the cycle-0 claim can never trigger a **WaterfallUnderflow** at runtime. The check is moved to the earliest possible moment — allocation — so an unviable pool never reaches `Active`.

3.1 Why 74% and not 100%

A naïve ROSCA would let 100% of each installment flow to the draw, maximizing the credit a given group can support. RoundFi deliberately gives up that headroom: the 26% it diverts (1% + 25%) is the capital that makes a default recoverable. The guard therefore encodes a **conservative** self-funding condition — the pool is solvent on the float **after** the shields have already been funded, not before. The trade is legible: a slightly smaller credit per installment in exchange for a pool that survives a defaulting member.

3.2 What the guard does and does not promise

The guard is a **cycle-0** invariant. It guarantees the pool can fund its first draw from contributions without subsidy, and — because `cycles_total == members_target` and each slot draws exactly once (MASTER-SPEC §4.1, the hard geometry invariant) — that the steady-state float arithmetic is closed. It does **not** by itself promise solvency through an arbitrary sequence of defaults; that is the job of the Triple Shield in §4. The two work together: the guard keeps the honest pool self-funding, and the shields backfill the float when a member fails to contribute.

4. The Triple Shield

When a member draws and then stops paying, the missed installment leaves a hole in the float — the pool still owes its **next** drawn member a full credit. The Triple Shield is the layered capital that fills that hole, seized in a fixed order on `settle_default` so that the cheapest, most communal buffer is spent first and the defaulter's own posted capital absorbs the rest.

4.1 The three layers

Layer	Capital	Funded by	Whose money
1	Solidarity vault	1% of every installment, all cycles	The pool's
2	Member escrow	25% of the defaulter's installments	The defaulter's
3	Member stake	Locked at join, sized by tier	The defaulter's

The progression is intentional. Layer 1 is **communal and small** — a shared first-loss buffer that smooths the common case (a single late installment) without reaching into anyone's collateral. Layers 2 and 3 are the **defaulter's own capital**: their vested-but-unreleased escrow, then their join stake. By the time the pool touches the stake, it has already exhausted the defaulter's other on-chain balances. Honest members' float is never the source of cover.

4.2 The seizure order

`settle_default(cycle)` draws cover in exactly this sequence (MASTER-SPEC §4.4):

1. **Solidarity vault** — up to the missed installment.
2. **Member escrow balance** — up to the remaining shortfall.
3. **Member stake** — the remainder.

All seized USDC flows into `pool_usdc_vault`, so the pool can still pay its drawn member. The defaulter is marked `defaulted = true` and a `SCHEMA_DEFAULT` attestation fires (the -500 score event of MASTER-SPEC §5.3).

The order is **solidarity** → **escrow** → **stake**, never the reverse. Seizing the shared buffer first means a recoverable, one-installment miss can be covered communally and cheaply; reaching the stake — the largest and most punitive layer — is reserved for the genuine shortfall that the first two layers couldn't close.

4.3 Worked stress cases

The cases below are illustrative arithmetic over the §4.4 seizure order and the §4.3 split; they are not new facts, only the model applied. Let a missed installment be I , the solidarity vault hold S , the defaulter's escrow balance E , and their stake K .

Case	Cover drawn	Touches honest float?
$S \geq I$	All from solidarity	No
$S < I \leq S + E$	Solidarity, then escrow	No
$S + E < I \leq S + E + K$	Solidarity, then escrow, then stake	No
$I > S + E + K$	All three layers, capped at total	Residual*

*The first three rows are the designed envelope: the missed installment is fully covered by the defaulter's posted capital plus the communal buffer, and the float is made whole. The fourth row is the **residual gap** the model is honest about — if a shortfall ever exceeded the entire Triple Shield, the layers cap at what they hold. The stake-sizing ladder (§6) exists precisely to keep that case out of reach: a member who can draw a large credit at a low stake is one whose long completed-pool history is itself the assurance they will not be in row four.

5. The vesting mirror — discipline releases what default seizes

Seizure is one half of the escrow surface; **vesting** is the mirror. The same escrow balance that backstops a default is also the member's own collateral, returned to them in tranches **as they prove on-time discipline**.

An on-time member progressively releases their stake from escrow via `release_escrow(checkpoint)`, gated on `member.on_time_count ≥ checkpoint` (MASTER-SPEC §4.4):

- **Pay on time** ⇒ the corresponding tranche unlocks and can be released.
- **Pay late** ⇒ the stake stays locked; `release_escrow` reverts with `EscrowLocked`.

This is the on-chain expression of "**discipline is rewarded, delinquency is collateralized.**" The two halves are economically symmetric:

Behavior	Effect on escrow
On-time payment	Advances <code>on_time_count</code> ; unlocks a vesting tranche
Late payment	No advance; stake remains locked (<code>EscrowLocked</code>)
Default	Escrow then stake seized into the pool float

The mirror is what makes the shield **fair as well as solvent**. A member who pays through to the end recovers their escrow and stake in full via on-time release; a member who defaults forfeits exactly that capital to the pool they failed. The collateral is never confiscated from the disciplined — it is released back to them on the same `on_time_count` signal that proves they earned it.

6. Stake economics — why the discount doesn't raise risk

The user-facing reward of the reputation engine is the **stake discount ladder**. A member stakes a fraction of the credit at join, sized by their tier (MASTER-SPEC §5.1):

Tier	Name	Stake (bps of credit)	Min completed pools
L1	Iniciante	5000 (50%)	0
L2	—	2500 (25%)	2
L3	Veteran	1000 (10%)	3
L4	Elite	300 (3%)	8

Source: MASTER-SPEC §5.1 / §8.3; `roundfi-reputation/src/constants.rs`
 (`STAKE_BPS_LEVEL_*` , `LEVEL_*_MIN_CYCLES`). The progression is **50** → **25** → **10** → **3**: an L4 Elite member frees roughly **94%** of the capital an L1 must lock.

6.1 The apparent paradox

Lowering the stake appears to weaken layer 3 of the Triple Shield — a defaulter at L4 forfeits only 3% of the credit, not 50%. Read naïvely, the discount looks like it trades solvency for a nicer incentive. It does not, and the reason is the gate.

6.2 The resolution — the discount is gated on the risk signal itself

The stake is not discounted for **claiming** a tier; it is discounted only after a member has **earned** it by completing pools to the end. The upper tiers are gated on a long completed-pool history (MASTER-SPEC §8.3):

- **L2** requires **2 completed pools** (raised from 1 in ECO-V52, so the 4× stake discount is never reachable on a single self-dealt pool).
- **L3** requires **3**.
- **L4** requires **8** — roughly a multi-year honest history.

Each `cycles_completed` increment fires only on a `POOL_COMPLETE` attestation — a pool paid **through to the end** — and carries a **30-day per-subject cooldown**

(`MIN_POOL_COMPLETE_COOLDOWN_SECS`). The history that unlocks the discount cannot be parallelized or bought; it accrues over wall-clock time, one completed obligation at a time.

So the protocol's risk **does not rise as the stake falls**, because the very condition that lowers the stake — a long record of completed pools — is itself the evidence that the member is unlikely to default. The stake is high precisely while a member is unproven (L1, 50%) and falls only as the behavioral case against default strengthens. The discount is **priced by demonstrated reliability**, not granted on entry.

6.3 The identity floor on the cheapest tier

The most-discounted tier carries the strongest additional wall: **L4 Elite always requires `identity_verified`**, independent of the configurable identity gate (MASTER-SPEC §5.2). The tier that risks the least locked capital is therefore also the one bound to proof-of-personhood — closing the path where an anonymous operator manufactures a low-stake position to default on. The economic prize and the strongest gate sit on the **same** tier by construction.

7. The yield cascade — surplus without subsidy

The solvency model is conservative by design (§3), which means idle float accumulates. Rather than let it sit, the pool can deposit idle float into a yield adapter (`deposit_idle_to_yield`) and harvest the surplus (`harvest_yield`). The realized surplus runs a fixed waterfall (MASTER-SPEC §8.2):

Order	Slice	Sizing	Nature
1	Protocol fee	<code>DEFAULT_FEE_BPS_YIELD</code> = 20% gross	Treasury — the only outflow
2	Guarantee Fund	logical earmark	<code>pool.guarantee_fund_balance</code>
3	LP slice	<code>config.lp_share_bps</code> (default 65%)	<code>pool.lp_distribution_balance</code>
4	Residual	remainder	Stays in <code>pool_usdc_vault</code>

The order is **20% → Guarantee Fund → 65% LP → residual**. Only the protocol fee is a **physical outflow**; the Guarantee Fund and LP slices are **logical earmarks** on pool balances, and the residual stays in the float as the participants' "prêmio de paciência" (patience premium). A slippage floor (`harvest_yield(min_realized)`) protects against an adapter under-reporting yield.

Two points matter for the solvency model. First, the **Guarantee Fund earmark** grows a pool-level reserve from surplus the conservative float threw off — a fourth, accumulating cushion behind the Triple Shield, funded by yield rather than by diverting contributions. Second, because the fee is the **only** outflow and the rest are earmarks, harvesting never moves capital out of the pool's solvency envelope except for that single, bounded protocol fee. Adapters are pluggable: the mock serves devnet, Kamino is the mainnet target.

8. Solvency invariants — the recoverability summary

Pulling the model together, a RoundFi default is recoverable because a small set of invariants hold simultaneously. Each is enforced in code at the earliest possible point, not assumed.

Invariant	Enforced by	Failure mode it removes
<code>members × installment × 0.74 ≥ credit</code>	<code>create_pool</code> → <code>PoolNotViable</code>	cycle-0 WaterfallUnderflow
<code>cycles_total == members_target</code>	<code>create_pool</code> (geometry invariant)	orphan cycles / unfunded draws
Seize solidarity → escrow → stake	<code>settle_default</code>	honest float absorbing a loss
On-time release only (<code>on_time_count</code>)	<code>release_escrow</code> → <code>Escrow-Locked</code>	premature collateral withdrawal
Default seizes before profile is portable	reputation keyed by subject = wallet	walking away with clean credit
Low stake gated on completed-pool history	<code>LEVEL_*_MIN_CYCLES</code> + 30-day cooldown	cheap-tier default farming

The chain reads cleanly: the **guard** makes the honest pool self-funding; the **Triple Shield**, seized in order, covers the defaulter's miss from the defaulter's own capital plus a communal buffer; the **vesting mirror** returns that same capital to members who stay disciplined; and the **gated discount** lowers the stake only as the behavioral case against default strengthens. No layer relies on trust where it could rely on an on-chain invariant — which is exactly the property a grant reviewer or actuarial reader needs to see before believing "recoverable" rather than merely hearing it.

Cross-references: protocol mechanics → [02-technical-whitepaper](#) ; program topology → [03-architecture-spec](#) ; the reputation ladder that prices the stake → [04-behavioral-reputation-score](#) ; adversarial economics → [09-risk-and-compliance](#) . Full economics live in MASTER-SPEC §4 and §8.